

SOC Task Report

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Role: SOC Intern

Platform: Windows 11 + Wazuh SIEM

Submission: SOC Fundamentals & Practical Tasks

1. Introduction

A Security Operations Center (SOC) is responsible for continuous monitoring, detection, investigation, and response to cybersecurity threats.

The objective of this task was to understand SOC fundamentals and perform basic security monitoring, log analysis, alert verification, and documentation using industry-relevant tools.

Date & Time	Source IP	Event ID	Description	Action Taken
23-Feb-2026	Localhost	4625	Multiple failed login attempts	Logged & monitored

2. SOC Fundamentals

The SOC operates to proactively detect threats, analyze alerts, and respond to incidents.

SOC roles include:

- Tier 1 Analyst – Initial alert monitoring and triage
- Tier 2 Analyst – Detailed investigation
- Tier 3 Analyst – Advanced threat handling
- SOC Manager – Oversees SOC operations
- Threat Hunter – Proactively searches for threats

Frameworks used in SOC include NIST Incident Response Framework and MITRE ATT&CK for mapping attacker techniques.

3. Tools Used

- Windows Event Viewer – Log analysis
- Wazuh SIEM – Security monitoring and alert visibility
- VirtualBox – Virtual lab environment
- Windows 11 – Endpoint log source

4. Log Analysis (Windows Event Viewer)

Windows Security logs were analyzed to identify failed login attempts.

Event ID 4625 was filtered to detect authentication failures.

Multiple failed login attempts were generated manually to simulate a brute-force attack scenario.

Logs were reviewed and exported in CSV format for documentation.

5. Security Monitoring Using Wazuh SIEM

Wazuh SIEM was deployed using a pre-configured virtual appliance.

The Wazuh Dashboard was accessed through a web browser.

Security logs were analyzed using the Explore → Discover section of the dashboard.

Authentication-related and system logs were visible, confirming that log collection and monitoring were functioning correctly.

6. Alert Detection and Monitoring

Default Wazuh authentication rules were used to detect failed login attempts.

Repeated authentication failures indicate possible brute-force attack behavior.

Alerts and events were verified through the Wazuh dashboard without creating custom rules, which reflects standard SOC practices.

7. Incident Response Workflow

Incident response follows the NIST lifecycle:

- Preparation
- Identification – Detection of multiple failed logins
- Containment – Monitoring or blocking suspicious IPs
- Eradication – Password reset or account lockout
- Recovery – Continued monitoring
- Lessons Learned – Improve authentication controls such as MFA

8. Documentation and Reporting

All observed events were documented using a structured incident template including:

Date/Time, Source IP, Event ID, Description, and Action Taken.

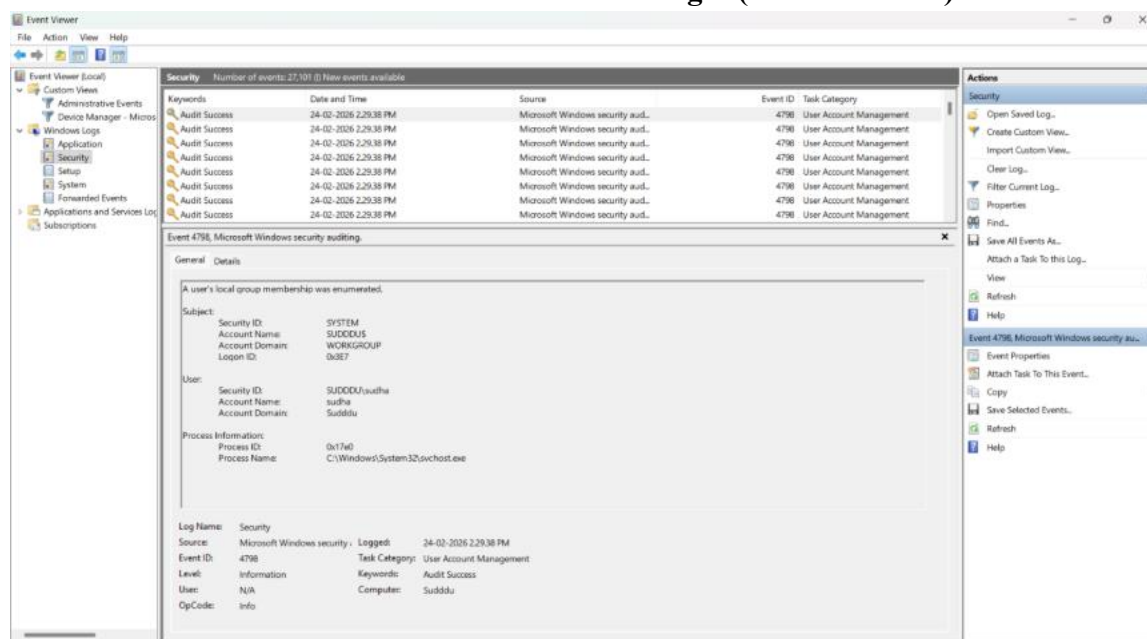
Screenshots and exported logs were maintained as evidence.

9. Conclusion

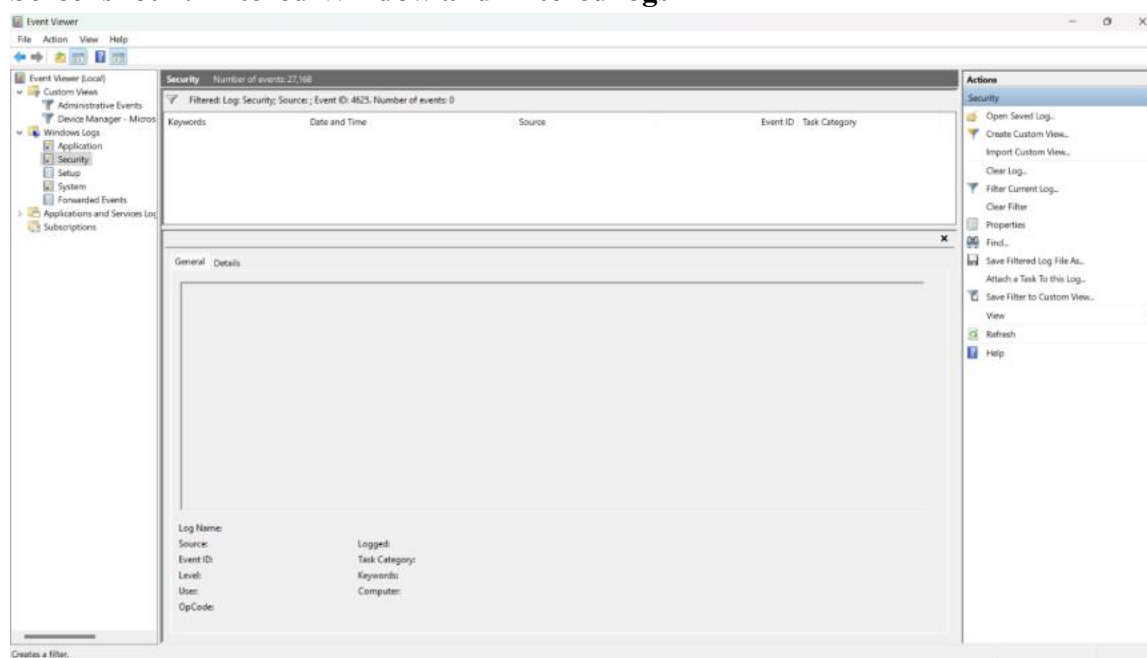
This task provided hands-on exposure to SOC operations, log analysis, and security monitoring. The exercise helped in understanding how SIEM tools assist SOC analysts in detecting and investigating security incidents.

Screenshot Evidence Section

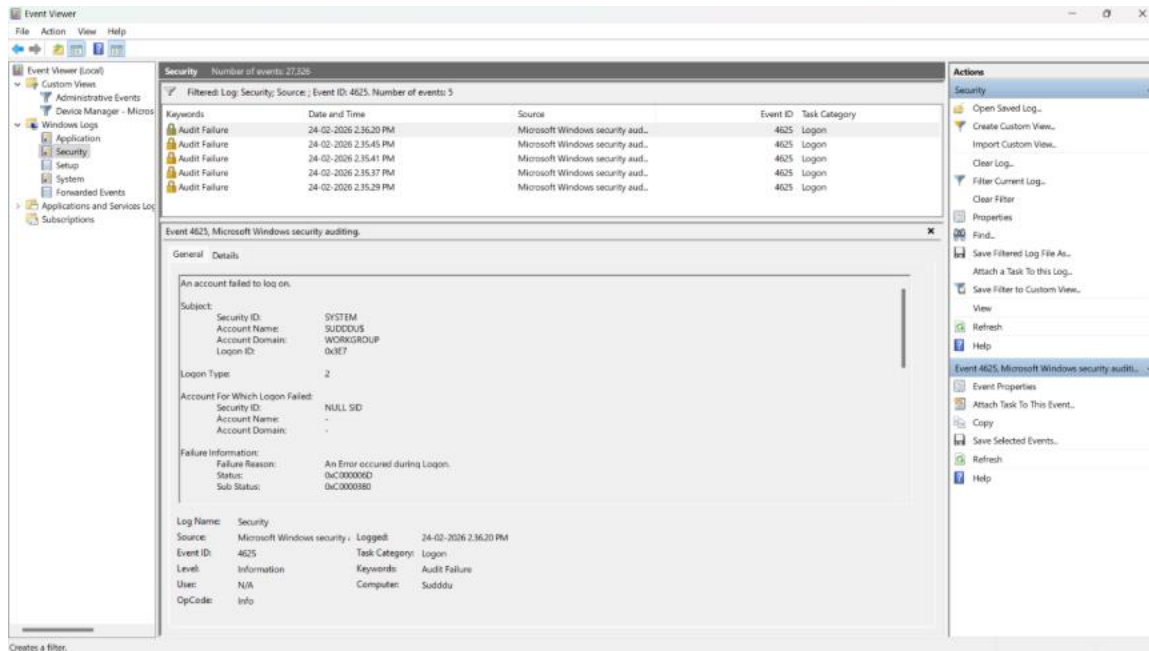
Screenshot 1: Windows Event Viewer – Failed Login (Event ID 4625)



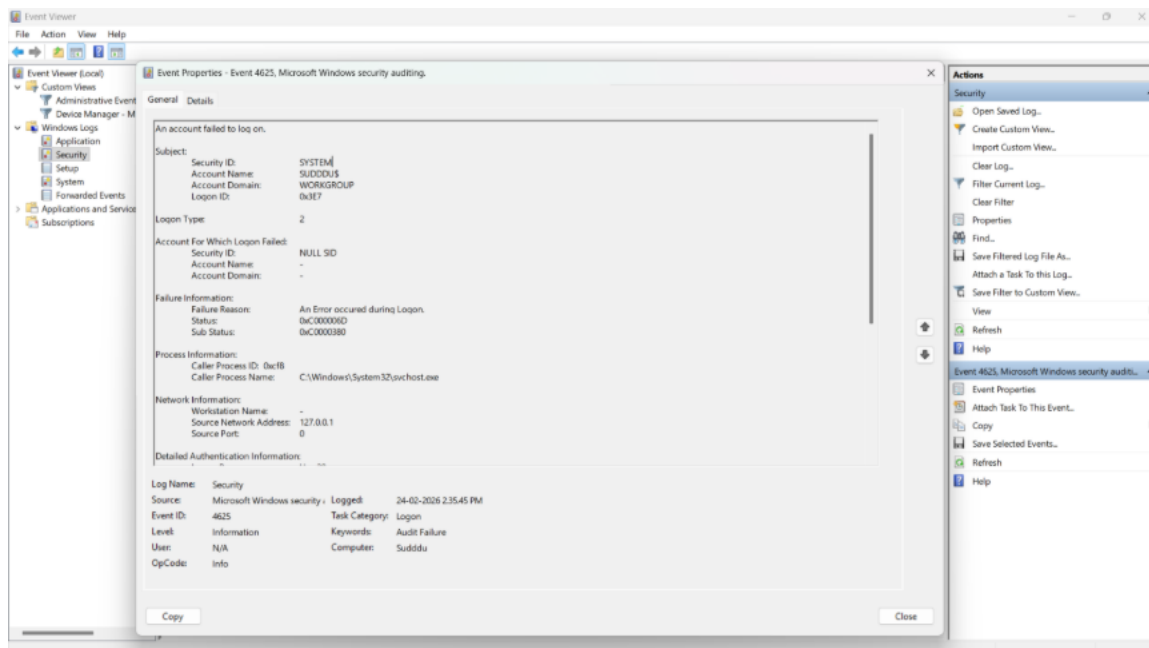
Screenshot 2: Filtered Window and Filtered logs



Screenshot 3: Failed login



Screenshot 4: Event Details Tab





Screenshot 5: Failed login

Event Viewer

File Action View Help

Event Viewer (local)

- Custom Views
- Administrative Events
- Device Manager - Micro
- Windows Logs
 - Application
 - Security
 - Setup
 - System
 - Forwarded Events
- Applications and Services Log
- Subscriptions

Security Number of events: 27,625

Filtered Log: Security; Source; Event ID: 4625. Number of events: 5

Keywords	Date and Time	Source	Event ID	Task Category
Audit Failure	24-02-2026 2:36:20 PM	Microsoft Windows security aud...	4625	Logon
Audit Failure	24-02-2026 2:35:45 PM	Microsoft Windows security aud...	4625	Logon
Audit Failure	24-02-2026 2:35:41 PM	Microsoft Windows security aud...	4625	Logon
Audit Failure	24-02-2026 2:35:37 PM	Microsoft Windows security aud...	4625	Logon
Audit Failure	24-02-2026 2:35:29 PM	Microsoft Windows security aud...	4625	Logon

Event 4625, Microsoft Windows security auditing

General Details

An account failed to log on.

Subject:

- Security ID: SYSTEM
- Account Name: SUDOUS
- Account Domain: WORKGROUP
- Logon ID: 0x07

Logon Type: 2

Account For Which Logon Failed:

- Security ID: NULL SID
- Account Name: -
- Account Domain: -

Failure Information:

- Failure Reason: An error occurred during Logon.
- Status: 0xC000006D
- Sub Status: 0xC0000080

Log Name: Security

Source: Microsoft Windows security - Logged: 24-02-2026 2:35:45 PM

Event ID: 4625 Task Category: Logon

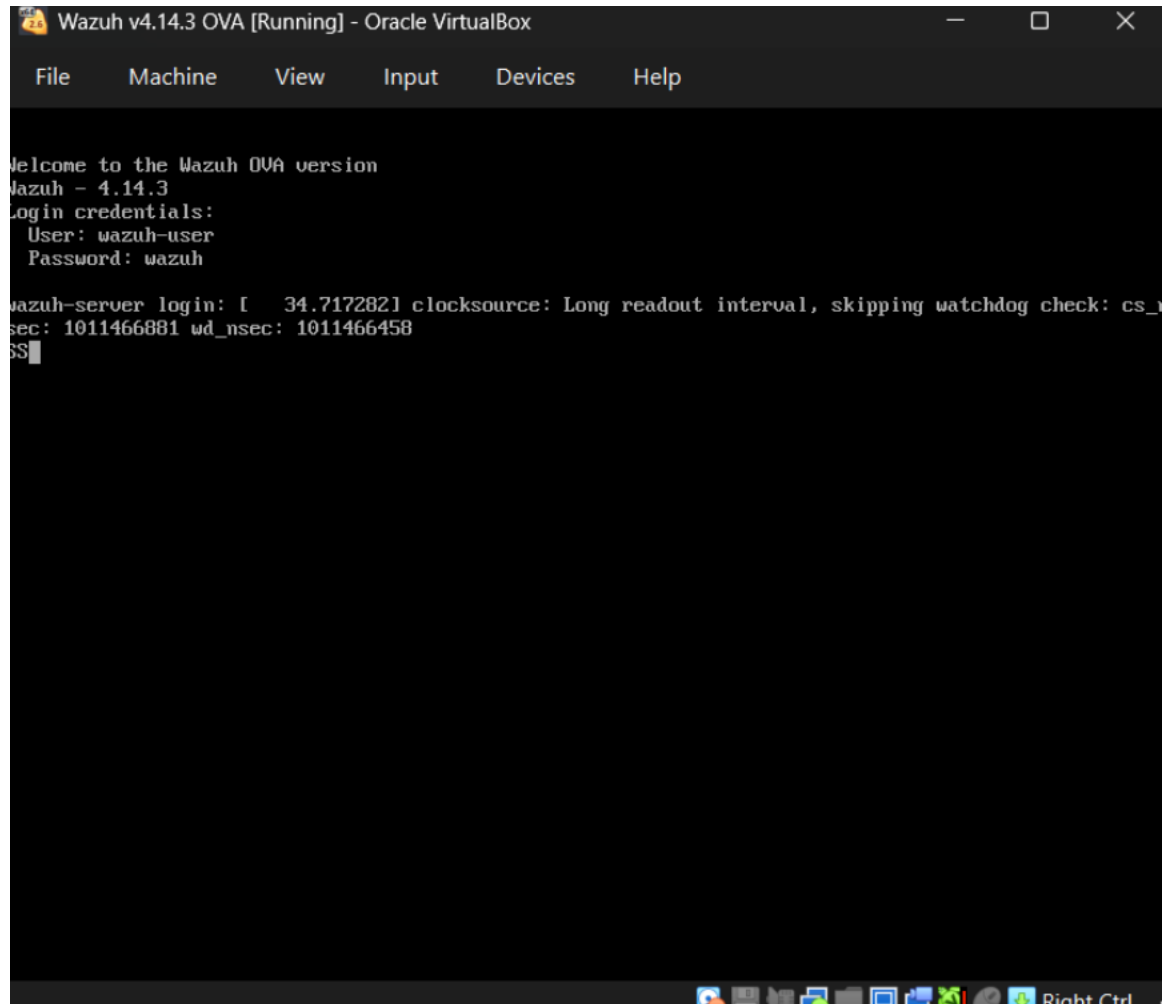
Level: Information Keywords: Audit Failure

User: N/A Computer: Suddu

OpCode: Info

Microsoft Management Console

Saving events to C:\MY_PC\CyArt Internship\SOC TASK \failed_logins_4625.csv' has completed successfully.

Screenshot 6: Wazuh in virtual box

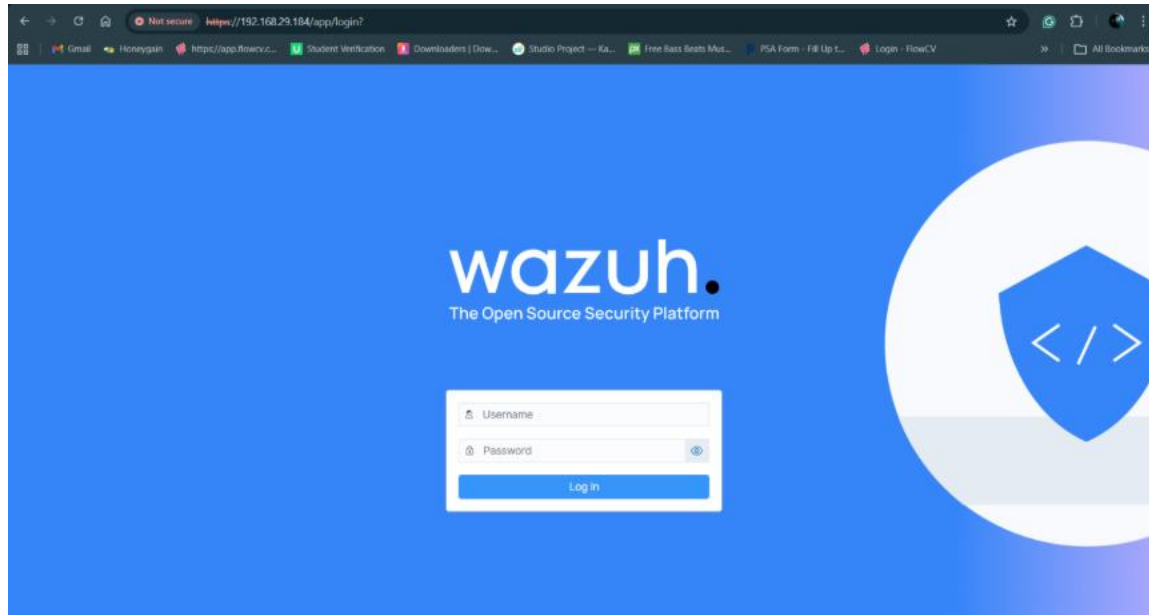
The screenshot shows a terminal window titled "Wazuh v4.14.3 OVA [Running] - Oracle VirtualBox". The terminal displays the following text:

```
Welcome to the Wazuh OVA version
Wazuh - 4.14.3
Login credentials:
  User: wazuh-user
  Password: wazuh

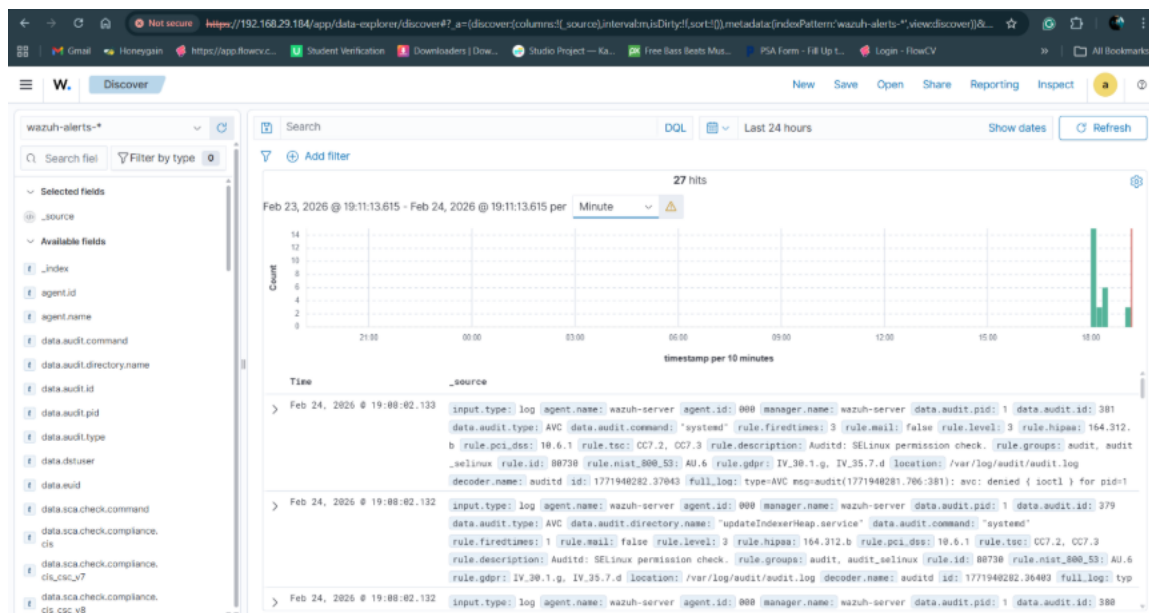
wazuh-server login: [ 34.717282] clocksource: Long readout interval, skipping watchdog check: cs_
sec: 1011466881 wd_nsec: 1011466458
SS
```

The terminal window has a menu bar with "File", "Machine", "View", "Input", "Devices", and "Help". The bottom of the window shows a taskbar with various icons and a "Right Ctrl" button.

Screenshot 7: Wazuh Dashboard



Screenshot 8: Alerts/Rules





Screenshot 9: Dashboard chart

